

# CampusGenie AI

## A Research Foundation Document

|           |                                                         |
|-----------|---------------------------------------------------------|
| Project   | CampusGenie AI – Agentic RAG Academic Assistant         |
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This document answers three core research questions:

- 1. What is the existing problem in university information access?
- 2. What research has been conducted and what does CampusGenie AI contribute?
- 3. What will be the impact and outcomes of this research?

All claims are supported by peer-reviewed references cited inline.

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# 1. The Existing Problem — What Is Wrong Today?

Universities are information-intensive organisations that generate and continuously update large volumes of institutional data — timetables, attendance records, examination notifications, placement circulars, hostel and transport notices, fee-payment deadlines, and policy documents governing eligibility, grading, and conduct. Despite the critical importance of this information to students and faculty, it remains fragmented across disconnected systems, creating a significant and measurable access problem.

■ Jurafsky, D., & Martin, J. H. (2023). *Speech and Language Processing* (3rd ed.). Pearson Education — establishes that information retrieval from heterogeneous, unstructured sources requires natural-language understanding layers that conventional portals do not provide.

## 1.1 Information Fragmentation

A student attempting to determine whether they meet the minimum attendance threshold to sit for an examination must consult an attendance dashboard, cross-reference it against a separate policy document specifying the threshold, and confirm the examination date from a third source — three independent lookups to answer what is, from the student's perspective, a single question. Multiplied across a student body of several thousand, the cumulative time cost becomes substantial.

- Timetables are stored separately from attendance systems.
- Examination notices are issued as PDF circulars, not linked to schedules.
- Placement eligibility criteria are maintained by a separate department.
- Fee schedules and hostel notices follow their own update cadences.
- Administrative regulations exist in policy documents not searchable by keyword.

## 1.2 Limitations of Conventional University Portals

Conventional university information systems are static: built around predefined navigation structures and keyword search, they cannot interpret the intent behind a naturally phrased question, cannot combine information drawn from more than one underlying source, and offer no mechanism for verifying that the information presented is current. These limitations are structural — they follow directly from the absence of any natural-language understanding or reasoning layer between the user and the underlying data.

■ Swacha, J., & Gracel, M. (2025). *Retrieval-Augmented Generation (RAG) Chatbots for Education: A Survey of Applications*. *AI*, 15(4234). MDPI — reviews 47 works (2023–2025) and confirms that most deployed university chatbots rely on narrow, hand-curated intent sets and rule-based response templates, making them rigid and unable to handle paraphrased or compound queries.

## 1.3 Why LLMs Alone Cannot Solve This

Large Language Models (LLMs), built on the Transformer architecture (Vaswani et al., 2017), now demonstrate strong natural-language understanding capability, making conversational interaction

over unstructured content technically feasible for the first time at scale. However, LLMs used in isolation are trained on static, general-purpose corpora and are consequently unable to answer questions about an institution's current notices, evolving policies, or private records.

■ Vaswani, A., et al. (2017). Attention Is All You Need. NeurIPS 30 — the foundational Transformer paper. LLMs built on this architecture have fixed parametric knowledge from their training corpus and no access to real-time institutional data.

When pressed for institution-specific information, LLMs are prone to hallucination — the confident generation of plausible but factually incorrect content. This is not a cosmetic weakness; an incorrect answer about an examination date or an eligibility threshold carries real consequences for the student who acts on it.

■ Huang, L., et al. (2023). A Survey on Hallucination in Large Language Models: Principles, Taxonomy, Challenges, and Open Questions. ACM Transactions on Information Systems. arXiv:2311.05232 — provides the definitive taxonomy of hallucination, distinguishing factuality hallucination (fabricated facts) from faithfulness hallucination (outputs inconsistent with provided context). Directly motivates retrieval-grounded design.

■ Tonmoy, S.M.T.I., et al. (2024). A Comprehensive Survey of Hallucination Mitigation Techniques in Large Language Models. arXiv:2401.01313 — surveys RAG, self-consistency, and chain-of-thought verification as the three primary mitigation families, confirming that retrieval-grounded generation is the most widely adopted and practically effective approach for domain-specific deployments.

1.4 Research Gap

| Dimension                      | LLM Chat Only | Rule-Based Chatbot | CampusGenie AI  |
|--------------------------------|---------------|--------------------|-----------------|
| Grounding in institution data  | Absent        | Static / Narrow    | Dynamic RAG     |
| Natural language understanding | Strong        | Weak               | Strong          |
| Multi-step reasoning           | Absent        | Absent             | Present (Agent) |
| Hallucination risk             | High          | Low but rigid      | Low (grounded)  |
| Extensibility                  | Low           | Very Low           | High (modular)  |

Table 1.1: Research gap — CampusGenie AI vs. existing approaches.

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## 2. The Research — What We Did and How

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CampusGenie AI integrates three mature research paradigms — Large Language Models, Retrieval-Augmented Generation, and Agentic AI orchestration — into a single, evaluable conversational system designed specifically for the university information-access problem. Each design decision is grounded in published research as detailed below.

### 2.1 Foundation: Retrieval-Augmented Generation (RAG)

The core architecture follows the Retrieval-Augmented Generation paradigm formalised by Lewis et al. (2020), in which an LLM's parametric knowledge is supplemented by a retrieval mechanism that fetches relevant, up-to-date passages from an external knowledge base at inference time. This allows a general-purpose LLM to be adapted to a specific, dynamic knowledge domain — such as a university's document repository — without any retraining of the underlying model, simply by updating the retrieval index.

■ Lewis, P., et al. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. NeurIPS 2020, 33, 9459–9474 — the foundational RAG paper. Formally defines the retrieve-then-generate architecture and demonstrates that grounding generation in retrieved external documents substantially reduces hallucination on knowledge-intensive tasks.

■ Gao, Y., et al. (2023). Retrieval-Augmented Generation for Large Language Models: A Survey. arXiv:2312.10997 — systematises RAG into Naive (single retrieve-then-generate pass), Advanced (query rewriting, re-ranking, iterative retrieval), and Modular (composable retrieval, routing, and generation modules) paradigms. CampusGenie AI is positioned as a Modular RAG system.

### 2.2 Dense Retrieval and Vector Search

Effective grounding depends on effective retrieval. CampusGenie AI represents both document chunks and incoming queries as dense vectors in a shared semantic embedding space, enabling semantic — rather than purely lexical — matching between a naturally phrased query and the institutional text that answers it.

■ Karpukhin, V., et al. (2020). Dense Passage Retrieval for Open-Domain Question Answering. EMNLP 2020 — demonstrates that learned dense vector representations substantially outperform sparse keyword-based retrieval (TF-IDF, BM25) for open-domain QA, particularly when queries are phrased differently from source text.

■ Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. EMNLP 2019 — proposes the Siamese network architecture for producing sentence-level embeddings efficiently, the standard building block of dense retrieval pipelines including our embedding model.

■ Johnson, J., Douze, M., & Jégou, H. (2019). Billion-Scale Similarity Search with GPUs. IEEE Transactions on Big Data — introduces FAISS, the approximate nearest-neighbour library used in CampusGenie AI's vector index for sub-second retrieval latency.

### 2.3 Agentic AI — The LangGraph ReAct Layer

A single retrieve-then-generate pass is insufficient for compound academic queries — for example, checking a student's attendance and then cross-referencing it against a separate eligibility policy before producing a synthesised answer. CampusGenie AI addresses this through an agentic orchestration layer built in LangGraph, following the ReAct reasoning-and-acting pattern.

■ Yao, S., et al. (2023). ReAct: Synergizing Reasoning and Acting in Language Models. ICLR 2023 — proposes interleaving explicit reasoning traces (thoughts) with tool-use actions and observations, allowing LLMs to plan, adapt, and self-correct across multiple steps. This is the exact pattern implemented in CampusGenie AI's LangGraph agent.

■ Wang, L., et al. (2023). A Survey on Large Language Model based Autonomous Agents. arXiv:2308.11432 — provides the unified profiling/memory/planning/action framework for LLM-based agents, situating ReAct as one instance of a much larger design space and informing the architecture of CampusGenie AI's agent layer.

■ Singh, A., et al. (2025). Agentic Retrieval-Augmented Generation: A Survey on Agentic RAG. arXiv:2501.09136 — the definitive 2025 survey of systems combining ReAct-style agents with RAG, covering Adaptive RAG, Corrective RAG, Self-RAG, and LangGraph-based implementations. CampusGenie AI is classified as an Agentic RAG system under this taxonomy.

■ Findings of IJCNLP 2025. Reasoning RAG via System 1 or System 2: A Survey on Reasoning Agentic Retrieval-Augmented Generation. ACL Anthology: 2025.findings-ijcnlp.122 — distinguishes predefined-pipeline RAG from agentic reasoning RAG where the model dynamically manages its own retrieval. The thought-action-observation loop in CampusGenie AI's LangGraph agent is classified as agentic reasoning under this taxonomy.

## 2.4 Advanced RAG Variants That Inform Our Design

### 2.4.1 Self-RAG — Selective Retrieval

Self-RAG introduces the idea that an LLM should decide whether retrieval is needed for a given query, rather than retrieving unconditionally. This directly informs the `classify_intent` node in CampusGenie AI's LangGraph workflow, which routes queries either to retrieval or directly to generation depending on query type.

■ Asai, A., et al. (2024). Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection. ICLR 2024 (Top 1% ranking) — introduces selective retrieval and self-reflection mechanisms. Adopted by LangChain and LlamaIndex. Best Paper Honorable Mention at NeurIPS 2023 Instruction Workshop.

### 2.4.2 CRAG — Corrective RAG

CRAG adds a lightweight retrieval evaluator that assesses the quality of retrieved documents before generation and triggers corrective actions — including a web search fallback — when retrieval confidence is low. Since CampusGenie AI uses LangGraph (which natively implements CRAG), this is a direct extension pathway for our system.

■ Yan, S.-Q., et al. (2024). Corrective Retrieval Augmented Generation. arXiv:2401.15884 — plug-and-play self-correction for RAG. Improves RAG accuracy by 15–37% across four benchmark datasets. Implemented natively in LangGraph, the exact framework used in CampusGenie AI.

### 2.4.3 Chunking Strategy

The quality of RAG grounding is bounded by how source documents are chunked before indexing. CampusGenie AI uses overlapping, semantically coherent chunks. Research confirms this is the right approach for documents combining structured data (timetables) and unstructured prose (policy

documents).

■ Merola, C., & Singh, J. (2025). Reconstructing Context: Evaluating Advanced Chunking Strategies for Retrieval-Augmented Generation. ECIR 2025 Workshop. arXiv:2504.19754 — rigorously compares late chunking vs. contextual retrieval vs. fixed-size chunking. Finds contextual retrieval preserves semantic coherence better but costs more; late chunking trades relevance for efficiency. Structure-aware chunking achieves highest F1 in generation by preserving paragraph logic.

## 2.5 Directly Comparable Prior Work

Several published systems are directly comparable to CampusGenie AI. Understanding these systems clarifies exactly where our contribution sits.

- Olawore, K., McTear, M., & Bi, Y. (2025). Development and Evaluation of a University Chatbot Using Deep Learning: A RAG-Based Approach. CONVERSATIONS 2024, Springer LNCS. DOI:10.1007/978-3-031-88045-2\_7 — uses LangChain, GPT-3.5/GPT-4, and FAISS for a university information chatbot. Closest published architecture to CampusGenie AI. Does NOT incorporate agentic multi-step reasoning — our key differentiator.
- Wibisono, Y., & Anisyah, A. (2025). Development of an Academic Services Chatbot Based on Retrieval-Augmented Generation (RAG). Brilliance, 5(2). DOI:10.47709/brilliance.v5i2.6719 — multi-stage RAG with query contextualization, reranking, and Gemini 2.5 Pro, validated explicitly with RAGAS. Most directly comparable in evaluation methodology.
- Nguyen, L., & Quan, T. (2025). URAG: Implementing a Unified Hybrid RAG for Precise Answers in University Admission Chatbots — A Case Study at HCMUT. ACM AIQAM '24. arXiv:2501.16276 — hybrid BM25 + FAISS retrieval deployed at a real university. Demonstrates that hybrid retrieval outperforms dense-only search for university queries, informing potential future enhancement of CampusGenie AI.
- Nguyen-Duc, A., et al. (2025). An Empirical Study of Multi-Agent RAG for Real-World University Admissions Counseling. arXiv:2507.11272 — presents MARAUS, combining hybrid FAISS + BM25 retrieval, multi-agent orchestration, and LLM-based generation for real university admissions in Vietnam. Validates our multi-agent design direction with real-world deployment evidence.
- Safonov, P. (2025). RAG-Based University Information Chatbot. Issues in Information Systems, 26(3), 413–420. IACIS — implements RAG over a university's private MySQL database (faculty profiles, lecture schedules) via a Telegram bot. Shows RAG working on structured relational university data, complementing our work on unstructured PDF documents.
- Lang, G., & Gurpinar, T. (2025). Retrieval-Augmented Generation (RAG) Chatbot Effectiveness in an Online Course. ISEDJ, 23(2). ERIC:EJ1467995 — empirical study from Quinnipiac University evaluating a RAG chatbot deployed in an actual online course with real student interaction data, providing external validation that RAG chatbots meaningfully improve student information access.

## 2.6 System Architecture Summary

| Layer        | Technology            | Research Basis                                 |
|--------------|-----------------------|------------------------------------------------|
| Presentation | React.js              | Component-based UI for streamed LLM output     |
| Application  | FastAPI (Python)      | Async I/O for concurrent LLM + vector DB calls |
| Agent        | LangGraph / LangChain | ReAct pattern — Yao et al. (2023)              |
| Retrieval    | ChromaDB / FAISS      | Johnson et al. (2019) — ANN vector search      |

|                   |                         |                                                     |
|-------------------|-------------------------|-----------------------------------------------------|
| <b>Embedding</b>  | Sentence-Transformers   | Reimers & Gurevych (2019) — SBERT                   |
| <b>Generation</b> | OpenAI API / Gemini API | GPT-4 (arXiv:2303.08774); Gemini (arXiv:2312.11805) |
| <b>Evaluation</b> | RAGAS Framework         | Es et al. (2024) — EACL System Demos                |

Table 2.1: System architecture with research basis for each layer.

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## 3. Impact — What This Research Will Achieve

The outcomes of CampusGenie AI can be understood at three levels: immediate practical impact for students and faculty, measurable technical contribution to the RAG and agentic AI literature, and long-term institutional value as an extensible platform.

### 3.1 Direct Impact on Students and Faculty

- Students will be able to query attendance, examination dates, placement eligibility, hostel policies, and fee schedules through a single conversational interface rather than navigating multiple disconnected systems.
- Faculty and administrative staff will experience reduced load from repetitive, factual queries — freeing time for higher-value academic interactions.
- Answers will be grounded in the institution's own, verifiable documents, eliminating the hallucination risk that makes general-purpose LLM chat unsuitable for institutional use.

■ Swacha, J., & Gracel, M. (2025). RAG Chatbots for Education: A Survey. MDPI AI 15(4234) — reviews 47 systems and confirms that RAG-based educational chatbots consistently reduce student information-search time and faculty query burden compared to static portals.

■ Lang, G., & Gurpinar, T. (2025). RAG Chatbot Effectiveness in an Online Course. ISEDJ, 23(2) — provides empirical evidence from a deployed RAG chatbot in a real university course, showing measurable improvement in students' ability to locate course information accurately.

### 3.2 Technical Contribution

CampusGenie AI makes several technical contributions relative to existing deployed university chatbot systems:

■ Contribution 1 — Agentic Multi-Step Resolution: Unlike prior university chatbots that perform a single retrieve-then-generate pass, CampusGenie AI implements a stateful LangGraph agent that can issue multiple retrieval calls across different document sources before synthesising a final answer. This directly addresses the compound query problem documented in the research gap analysis.

■ Contribution 2 — Modular, Extensible Architecture: The six-layer design (Presentation → Application → Agent → Retrieval → Data → Generation) allows new capabilities — voice input, document summarisation, assignment assistance — to be added as agent tools without redesigning the core system.

■ Contribution 3 — Rigorous Component-Level Evaluation: Using RAGAS metrics (context precision, context recall, faithfulness, answer relevance) applied to a held-out query set spanning simple lookups, compound queries, and out-of-scope queries — a level of evaluation discipline absent from most existing university chatbot deployments.

■ Es, S., et al. (2024). RAGAs: Automated Evaluation of Retrieval Augmented Generation. EACL 2024 System Demonstrations, 150–158 — introduces the reference-free evaluation framework adopted in this project. The four metrics (context precision, context recall, faithfulness, answer relevance) allow retrieval quality and generation quality to be diagnosed separately, enabling iterative improvement.

■ Yu, H., et al. (2024). Evaluation of Retrieval-Augmented Generation: A Survey. arXiv:2405.07437 — surveys all major RAG evaluation frameworks (RAGAS, ARES, RAGCHECKER, RECALL) and confirms

RAGAS as the most widely adopted, reference-free, component-level evaluation suite for RAG systems.

### 3.3 Future Scope Enabled by This Research

Because CampusGenie AI is designed as an extensible, modular platform, the research opens clear pathways for subsequent work:

- Voice interaction: extending the presentation layer with speech-to-text and text-to-speech so queries can be spoken rather than typed.
- GraphRAG integration: representing relationships between university entities (student → course → faculty → department → policy) as a knowledge graph, enabling richer multi-hop reasoning beyond flat vector retrieval.

■ Edge, D., et al. (2024). GraphRAG: Unlocking LLM discovery on narrative private data. Microsoft Research Blog, July 2024 — GraphRAG uses LLM-generated knowledge graphs to improve retrieval for queries requiring connection of multiple related entities, a natural extension for university knowledge bases where entity relationships are structured and predictable.

- Hybrid BM25 + dense retrieval: adding keyword-based retrieval alongside semantic search to improve precision for structured institutional data like timetables and fee schedules.

■ Nguyen, L., & Quan, T. (2025). URAG: Unified Hybrid RAG. ACM AIQAM '24. arXiv:2501.16276 — demonstrates that combining BM25 keyword retrieval with FAISS dense retrieval, re-ranked by a cross-encoder, substantially outperforms dense-only retrieval for university information queries.

- Corrective RAG (CRAG) integration: adding a lightweight retrieval evaluator that triggers corrective retrieval or web search when retrieved context confidence is below threshold.

■ Yan, S.-Q., et al. (2024). CRAG. arXiv:2401.15884 — implemented natively in LangGraph, the exact orchestration framework used in CampusGenie AI, making CRAG integration a low-friction next step.

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| Nguyen-Duc et al. 2025 (MAR AUS)        | <a href="https://arxiv.org/pdf/2507.11272">arxiv.org/pdf/2507.11272</a>                                                           |
| Yu et al. 2024 (RAG Eval Survey)        | <a href="https://arxiv.org/pdf/2405.07437">arxiv.org/pdf/2405.07437</a>                                                           |
| Es et al. 2024 (RAGAS)                  | <a href="https://aclanthology.org/2024.eacl-demo.16.pdf">aclanthology.org/2024.eacl-demo.16.pdf</a>                               |
| Jurafsky & Martin (textbook)            | <a href="https://web.stanford.edu/~jurafsky/slp3/">web.stanford.edu/~jurafsky/slp3/</a>                                           |
| Wibisono & Anisyah 2025                 | <a href="https://doi.org/10.47709/brilliance.v5i2.6719">doi.org/10.47709/brilliance.v5i2.6719</a>                                 |
| Safonov 2025 (IACIS)                    | <a href="https://iacis.org/iis/2025/3_iis_2025_413-420.pdf">iacis.org/iis/2025/3_iis_2025_413-420.pdf</a>                         |
| Lang & Gurpinar 2025                    | <a href="https://files.eric.ed.gov/fulltext/EJ1467995.pdf">files.eric.ed.gov/fulltext/EJ1467995.pdf</a>                           |
| Swacha & Gracel 2025 (MDPI)             | <a href="https://mdpi.com/2673-2688/6/1/15">mdpi.com/2673-2688/6/1/15</a>                                                         |
| Olawore et al. 2025 (Springer)          | <a href="https://link.springer.com/chapter/10.1007/978-3-031-88045-2_7">link.springer.com/chapter/10.1007/978-3-031-88045-2_7</a> |
| GraphRAG — Microsoft Research           | <a href="https://microsoft.com/en-us/research/project/graphrag/">microsoft.com/en-us/research/project/graphrag/</a>               |

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